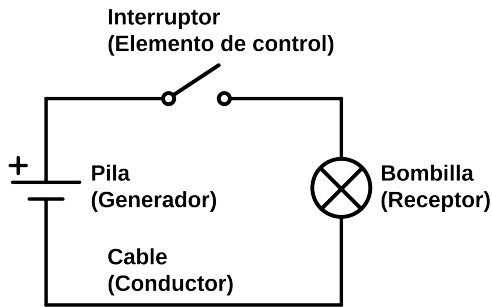


THE ELECTRICAL CIRCUIT

1. The electrical circuit.

An electrical circuit is a set of components that generate and control the flow of electricity to produce useful effects.



A simple example of an electrical circuit is the one we all use when we turn on the light in a room.

Circuits are made up of four types of components: generators, conductors, receivers, and control elements. Each of them will be studied in more detail below.

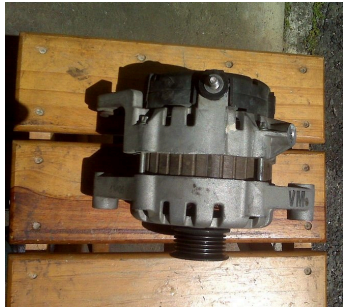
2. Generators

These components are responsible for generating electric current. To do this, they push electrons so that they flow through the circuit.

Examples of generators are cells and batteries, bicycle dynamos, car alternators, and photovoltaic solar panels.



Electric batteries.
Lead Holder, CC BY-SA 3.0



Car electric alternator.
El Guarito, CC BY-SA 3.0



Photovoltaic panel for solar electricity generation.
Saintfevrier, Public domain

3. Conductors

Conductors carry electricity between the components of the circuit. They are usually electrical wires.

The most common materials used to conduct electricity are:

Copper:

It is the most widely used material inside buildings, in the flexible wires of appliances, and for manufacturing electric motors.

Aluminum and steel:

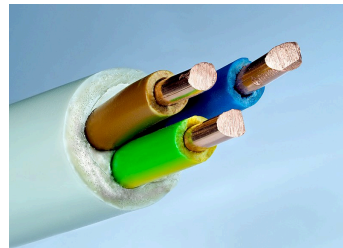
They are the most widely used materials in high-voltage cables. They have good mechanical strength, resist oxidation well, and are cheaper than copper.

Gold, nickel, and chromium:

They are used to coat electrical contacts to prevent oxidation and improve conductivity. They can be found in audio plugs and USB connectors.

Tin, lead, and silver:

Because of their low melting temperature (below 300°C), they are used in the soldering of electronic components. Silver, despite being more expensive, is increasingly used because it does not produce the toxic effects associated with lead.



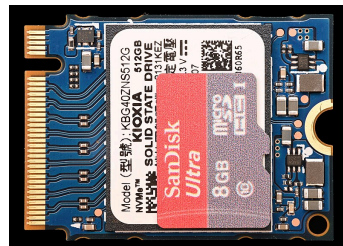
Copper cable with 3 wires
with a cross-sectional area
of 2.5mm².

Petar Milošević, CC BY-SA
4.0



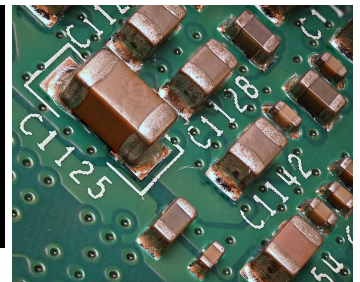
High-voltage cable made of
aluminum and steel.

Albarubescens, CC BY-SA
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SSD drive with gold-plated
connectors.

Phiarc, CC BY-SA 4.0



SMD components attached
to the PCB with tin-lead solder.

Phiarc, CC BY-SA 4.0

4. Receivers

Receiver components transform electricity into useful effects such as light, heat, movement, sound, etc.

Some examples of receivers are light bulbs, which produce light, motors, which produce movement, resistors, which produce heat, or bells, which produce sound.



LED lamp. Produces light from electricity.
Mcapdevila, CC BY-SA 3.0



Electric heating element in a ceramic cooktop. Produces heat.
A. Savin, CC BY-SA 3.0

5. Control elements

These elements make it possible to control the flow of electricity as needed. The simplest example is a switch that turns a light on or off by allowing electricity to flow when desired.

Depending on how they are operated, there are several types of control elements.

Manual operation:

Switches, push buttons, rotary controls, etc. They allow people to control electrical appliances.

Each manual control element has its practical application. When controlling a bell, a switch cannot be used because after pressing it, the bell would continue to operate without stopping. In this application, it is better to use a push button, which only activates the bell while it is being pressed.

Electrical protection:

Fuses, circuit breakers, residual-current devices, etc.

Fuses and **circuit breakers** interrupt the electric current to protect the electrical installation and prevent the wires from burning out in the event of a short circuit or overload.

The **residual-current device** protects people's lives by interrupting the electric current before a fault current can cause an electric shock.

Automatic operation:

Some control elements are operated by electrical signals. This allows automatic control, reducing the need for human intervention.

Examples of automatic operation include: a staircase light that switches off by itself after a certain amount of time, an electric door that opens automatically when it detects someone's presence, an elevator that stops on the correct floor thanks to a limit switch, and a smart building that uses a computer to control temperature, humidity, the opening of blinds, automatic irrigation, etc.



On/off switch.
Jszack, CC BY-SA 2.5



Residual-current device.
Protects people from electric shocks.
Raimond Spekking, CC BY-SA 4.0

EXERCISES

1. Draw a simple electrical circuit for a flashlight that turns on a lamp with a switch. Write the name of its components and what type each one is.
2. What is an electrical generator and what is it used for? Write three different examples of electrical generators.
3. What is an electrical conductor and what is it used for?
4. Write four different types of conductors and what each one is used for.
5. What function do receivers have in an electrical circuit?
6. Write four different receivers and the effect produced by each one.
7. What function do control elements have in an electrical circuit?
8. Write an example of a manual control element, a protection element, and an automatic control element.